Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Protection of miniaturized or flexible electronics in harsh environments			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		**Mechanical・Materials**	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	*Bio-signal Measurement / Diagnostic Devices*	*Nano/Micro Mechanical Systems*	*System Integration Technologies*
Concept of Proposed Technology	Core concept: The project aims to develop new material and structural concepts for protecting miniaturized and flexible electronics in mechanically, thermally, or chemically demanding environments. The focus is on ultra-thin protective layers and adaptive encapsulation technologies that ensure both high flexibility and long-term stability. By combining nanostructured barrier materials, innovative coating processes (e.g., ALD, parylene, or nanograded multilayers), and functionally integrated substrates, the sensitivity of flexible electronics to moisture, corrosion, mechanical stress, and temperature cycles is to be significantly reduced. Main functions: Mechanical protection: Increased bending and torsional resistance thanks to elastic, multi-layer protection systems. Chemical and thermal stability: Reduced penetration of water, oxygen, or aggressive media; stability at elevated temperatures. Self-monitoring and adaptivity: Integration of sensor interfaces or self-healing polymer layers for early detection and			

Rationale for Joint R&D Project	repair of microdamage. Process compatibility: Compatibility with established thin-film and roll-to-roll manufacturing processes for scalability of the technology. Potential applications: The protective concepts developed open up a wide range of application prospects in various high-tech fields: Medical technology: Protection of flexible bioelectronic sensors or implants against body fluids and mechanical stress. Industry 4.0 and IoT: Long-term functionality of sensors in industrial or maritime environments. Aerospace: Use in lightweight, structurally integrated electronic systems with high temperature and vibration requirements. Wearables and smart textiles: Improvement of the service life and reliability of flexible systems in direct skin contact. Energy and environmental technology: Protection of thin-film solar cells or flexible battery systems in outdoor applications.
	Necessity of the joint research and development project and rationale for cooperation The increasing prevalence of miniaturized and flexible electronic systems is opening up new perspectives for applications in medical technology, manufacturing automation, energy conversion, and wearable technologies. At the same time, today's protection and packaging technologies are reaching their limits when it comes to ensuring that these systems function reliably over the long term under harsh environmental conditions, such as high humidity, chemical exposure, temperature changes, or mechanical deformation. Purely national or monodisciplinary development cannot adequately meet these complex requirements, as there is a lack of integrated approaches that take materials, processing, structural mechanics, and system integration into account in equal measure. The joint R&D initiative is therefore necessary in order to: Pool material and process expertise – for example, through cooperation between the Steinbeis Transfer Center for Surface and Coating Technology (with expertise in material characterization, barrier technologies, and reliability analysis) and Korean industry partners (with experience in large-scale manufacturing, system integration, and product validation). Accelerate technology transfer and commercialization – through the close involvement of small and medium-sized enterprises (SMEs) that can translate the developed protection concepts into real product solutions. Harmonize standards and test protocols internationally to enable the market introduction of flexible electronic systems with high environmental stability. Realize synergies in research and application, in particular by utilizing complementary laboratory infrastructures, material platforms, and testing technologies of both partner countries. Cooperation is essential because it brings together interdisciplinary expertise that individual institutions alone can hardly cover. While the Steinbeis Transfer Center is primarily a leader in applied research on material modification, protective layer design, and reliability analysis, Korean partners have direct access to industrially relevant manufacturing environments and validation fields through TIPA-mediated networks. This creates a closed innovation chain from materials to components to marketable products. In addition, the project supports TIPA's strategic goals – namely, promoting technological innovation in SMEs, establishing international R&D networks, and accelerating the commercialization of innovative technologies that strengthen the competitiveness of industry in both countries.
Key Words	1. Diffusion Barrier 2. Atomic Layer Deposition ALD

	3. Parylene 4. Harsh Environment 5. Flexible Electronics

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3-4)	**End of the Project**	Level (5-6)
R&D Objectives	Depending on the planned areas of application for miniaturized electronics and sensor technology, application-specific encapsulation concepts are to be developed on a laboratory scale and tested in storage tests.			
R&D Target Specifications	The service life required for a system in harsh environments (e.g., for implants in physiological environments for short-term or long-term implantation) should be achieved through suitable multi-layer coatings. These specifications should be verified in appropriate aging tests. <u>Key research questions</u> • How can flexible substrates, conductor tracks, and connections be protected so that they retain their function in harsh conditions? • Which materials and protective coatings offer the optimal compromise between flexibility, barrier effect, and thermal stability? • Which test methods and aging tests are suitable for simulating the real operating environment?			

R&D Plan	• WP1: Requirements analysis & specification:
	  ○ Definition of operating conditions, determination of electrical, thermal, and mechanical requirements for flexible systems (all)
	• WP2: Material selection & development
	  ○ Selection and synthesis of protective coatings (polymers, nanocomposites, flame and moisture protection), characterization of chemical stability (Steinbeis)
	• WP3: Coating and encapsulation technologies
	  ○ Development and optimization of thin film or multilayer systems (e.g., via ALD, PECVD, Parylene-deposition, …t.b.d.) (Steinbeis, industry partners Korea)
	• WP4: Integration on flexible electronics platforms,
	  ○ Application of protection systems to real components (sensors, conductor tracks, microcontrollers), (Steinbeis, industry partners Korea)
	• WP5: Environmental simulation & reliability testing,
	  ○ Conducting temperature, humidity, vibration, and chemical tests (Steinbeis)
	• WP6: Evaluation, optimization & scaling
	  ○ Technical-economic evaluation, preparation of production and transfer concepts (all)
	• WP7: Project management & dissemination
	  ○ Coordination, documentation (Steinbeis)
	  ○ Publication results, workshops (all)			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The scientific partner takes the leading role in the fundamental research, development, and characterization of new material and structural concepts for the protection of miniaturized and flexible electronic systems. The institute contributes its extensive expertise in materials science, thin-film technologies, nanostructuring, and reliability analysis. The key responsibilities of the scientific partner include: • Material and process development: Investigation and optimization of ultra-thin barrier and protective layers based on nanostructured and functionally graded materials. • Development and adaptation of coating processes: Research on innovative coating technologies (e.g., Atomic Layer Deposition (ALD), parylene deposition, nanograded multilayer systems) and their transferability to flexible substrates. • Structural and physico-chemical characterization: Analytical evaluation of the developed materials regarding layer thickness, morphology, adhesion, thermal stability, and resistance to moisture and corrosion. • Mechanical and environmental testing: Experimental assessment of mechanical durability under bending, tensile/compressive stress, thermal cycling, and humidity exposure. • Scientific coordination and knowledge transfer: Coordination of the scientific work packages, evaluation of results, and transfer of research findings to industrial partners in Korea for technological implementation. Through these activities, the scientific partner provides the scientific and technological foundation for the development of innovative and application-oriented protective and encapsulation concepts, thereby supporting the industrial utilization of the research results by the Korean partners.
Expected R&R of Korean SME Partner	The Korean industrial partner focuses on the technological implementation, scaling, and application-oriented validation of the materials and processes developed within the project. The company contributes its industrial expertise in flexible and miniaturized electronic systems, as well as in device integration and product reliability. The key responsibilities of the industrial partner include: • Industrial process integration: Adaptation and incorporation of the newly developed protective layers and encapsulation technologies into existing or emerging manufacturing processes for flexible electronic components. • Prototype fabrication and validation: Production of demonstrator devices and prototypes using the developed materials and coating methods, followed by evaluation under real-world operational conditions. • Performance and reliability testing: Execution of standardized industrial tests (e.g., thermal cycling, bending, moisture exposure) to assess the long-term stability, durability, and manufacturability of the developed protection concepts. • Feedback for material and process optimization: Providing continuous feedback to the scientific partner based on industrial testing results to refine material formulations and coating procedures. • Pathway to commercialization: Evaluation of scalability, cost efficiency, and potential industrial applications, as well as preparation for technology transfer and market introduction. By performing these tasks, the Korean industrial partner ensures the practical validation and industrial relevance of the project outcomes, bridging the gap between fundamental research and commercial applications.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technological Achievements The project is expected to generate substantial advancements in the protection of miniaturized and flexible electronics operating under demanding mechanical, thermal, and chemical conditions. Key technological achievements will include: Development of ultra-thin, multifunctional barrier layers that provide outstanding moisture and corrosion resistance while maintaining high flexibility. Novel adaptive encapsulation strategies capable of responding to mechanical deformation and environmental fluctuations without compromising device performance. Integration of advanced coating technologies such as Atomic Layer Deposition (ALD), parylene deposition, and nanograded multilayer fabrication for improved process control and uniformity on flexible substrates. Comprehensive characterization database and reliability assessment that will serve as reference for future industrial adoption and standardization. 2. Commercialization Potential The outcomes of the project will open new pathways for commercial applications in the rapidly growing sectors of flexible and wearable electronics, sensors, medical devices, and next-generation packaging technologies. The developed materials and coating processes will enhance the durability and lifecycle of flexible devices, thus increasing their competitiveness and market value. The Korean industrial partner will leverage its production capabilities to scale up the most promising materials and processes, paving the way for prototype production and pilot manufacturing. Opportunities for joint intellectual property (IP) generation and technology transfer will further strengthen the industrial partnership between Korea and Germany. 3. Industrial and Economic Impact Beyond direct technological outcomes, the project will have a broad impact on both the industrial and economic levels: It will foster cross-border innovation and knowledge exchange between Korean industry and German research excellence in advanced materials science. The project will contribute to the competitiveness of both nations' flexible electronics industries, positioning them as leaders in high-performance protective solutions. Long-term, the developed technologies are expected to reduce production costs, improve product reliability, and expand the application fields of flexible electronics in areas such as healthcare, automotive, and Internet of Things (IoT) systems. 4. Broader Scientific and Societal Benefits • Strengthened collaboration networks between academia, research institutes, and industry across Germany and Korea. • Advancement in eco-efficient and sustainable material concepts, contributing to the green transition in electronics manufacturing. • Training and capacity-building of young researchers and engineers in cutting-edge thin-film and nanomaterial technologies.

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b) Any other form of publicly disclosed results

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization